Array Geometry Comparison: Theoretical Screening Results

IYPT Pinhole Sunglasses Project — Array Design Sub-Study

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1 Purpose and Scope

This report consolidates the theoretical (closed-form) screening comparison of nine pinhole-array geometry families: square, hexagonal, rectangular (anisotropic), concentric rings, ring-offset concentric, Fibonacci spiral, golden-angle phyllotaxis, Archimedean spiral, and two disordered families (Poisson-disk, jittered lattice).

At this stage, hole diameter a_{hole} is *fixed* per myopia level M from the project's existing acuity-optimal aperture chart (`new_sweeps/per_M_optimum_map.csv`) and is not re-optimized here. This study evaluates only geometry (pattern, hole count N, spacing) against two independently validated closed-form physical constraints: ghost-image separation and gaze-angle vignetting. No brightness/transmission objective is used in this pass (explicit scope decision — see Section 5).

2 Data Integrity

A degeneracy defect was found and corrected during this study: severe overlap-rejection collapse (requested pitch smaller than hole diameter) produced candidate rows where the per-family area formula no longer represented a valid periodic tiling or convex hull, yielding physically impossible packing-density values (open-area fraction > 1). This was diagnosed to two independent mechanisms — a spacing-independent unit-cell infeasibility for lattice families, and a boundary hole-radius omission for hull/circle-based families — and fixed at the root of the dataset, not patched downstream. The exclusion logic (`degeneracy_filter.py`) is now the single shared source of truth used by every downstream script.

Quantity	Value
Raw rows generated (pre-cleanup)	6615
Degenerate rows identified and excluded	0
Active rows in sweep_results.csv (post-cleanup)	6395
Geometry families tested	10
Physically-impossible density rows remaining (open area fraction > 1)	0

Exclusion reasons (220 rows total, grouped):

Exclusion reason (truncated)	Count

3 Safety Gates

Two independent, composable gates are applied to every candidate row:

- **Gate 1 (ghost safety):** worst-case ghost-separation margin > 0, using the empirically validated formula $\Delta r = sM/(P_0 - M)$ ($R^2 = 0.995$ against direct two-hole coherent wave-optics propagation).

- **Gate 2 (vignetting safety):** worst-case margin > 0 across gaze angle $\varphi \in [-25°, 25°]$, using the validated Handbook 2 Ch. 13 circle-overlap geometry.

A third gate (seed-reproducibility, for the two stochastic families) was considered and explicitly *not* applied in this pass, by decision: a stochastic family's greater susceptibility to failing Gates 1–2 is intended to show up naturally in survivor counts rather than be pre-filtered.

3.1 Survivor Funnel per Family

Family	Total rows	Ghost-safe (Gate 1)	Both gates	% survive
archimedean spiral	135	66	26	19.3%
concentric	318	110	73	23.0%
fibonacci spiral	133	62	14	10.5%
golden angle	133	62	14	10.5%
hexagonal	132	60	33	25.0%
jittered lattice	3792	1725	663	17.5%
poisson disk	864	304	180	20.8%
rectangular	432	191	113	26.2%
ring offset concentric	318	112	65	20.4%
square	138	60	35	25.4%

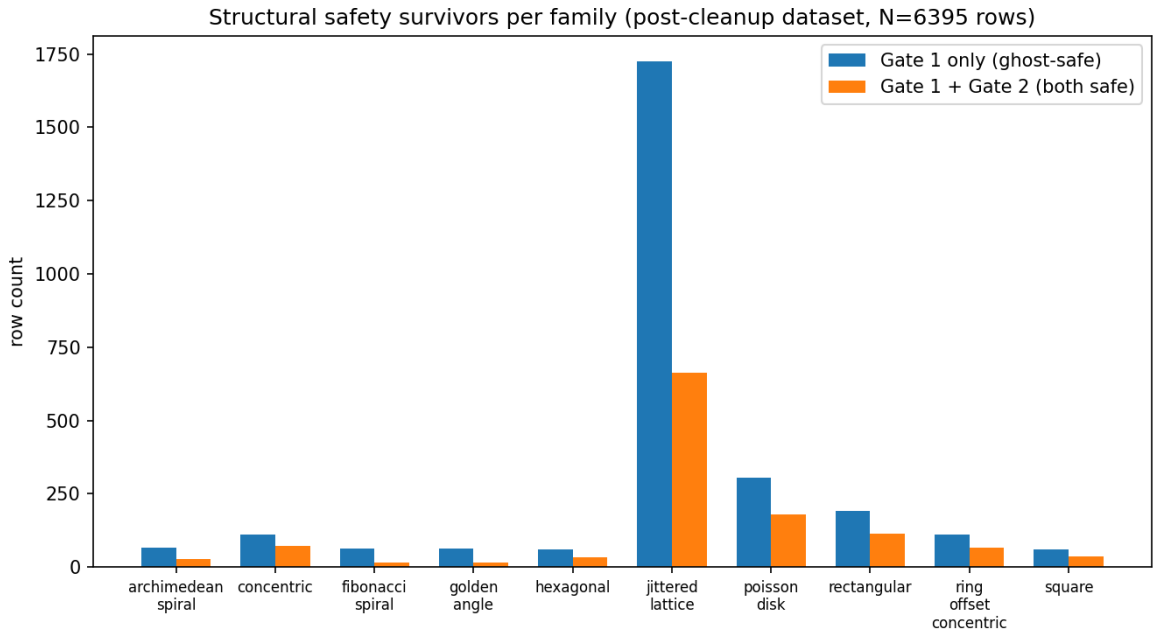


Figure 1: Rows surviving Gate 1 alone versus both gates, per family, across the full cleaned dataset (6395 rows, 9 families).

3.2 Ghost/Vignetting Divergence

A central finding of this study: passing Gate 1 with an infinite margin (i.e. no hole pairs simultaneously active under the pupil at $\varphi = 0$) does *not* imply Gate 2 is also satisfied. The two criteria must be evaluated independently for every family — one cannot be used as a proxy for the other.

Family	ϕ=0 ghost-safe rows	Also vignetting-safe	Fails vignetting	Divergence
concentric	98	61	37	37.8%
rectangular	190	112	78	41.1%
poisson disk	302	178	124	41.1%
square	60	35	25	41.7%
hexagonal	60	33	27	45.0%
ring offset concentric	102	55	47	46.1%
archimedean spiral	66	26	40	60.6%
jittered lattice	1690	628	1062	62.8%
fibonacci spiral	54	6	48	88.9%
golden angle	54	6	48	88.9%

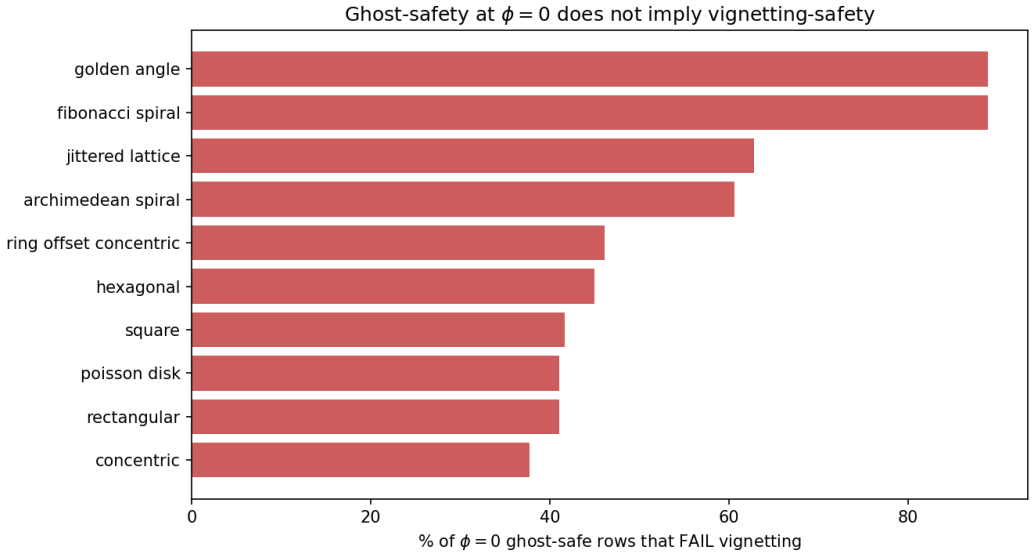


Figure 2: Fraction of φ=0 ghost-safe candidates that fail vignetting safety once gaze angle is integrated, per family. Divergence ranges from 37.8% (concentric) to 88.9% (Fibonacci spiral / golden angle).

4 Comparison Metrics

Two metrics are computed over the Gate 1∩Gate 2 survivor set:

Metric 1 — maximum packing density. Per M (and a global, worst-case-across-M scope), the single surviving candidate with highest N_{actual} per unit packed area.

Metric 2 — Pareto front (density vs. binding margin). The full 2D dominance front over (packing density, min(ghost margin, vignetting margin)), showing the trade-off rather than collapsing to one point.

4.1 Metric 1 Winners (post-cleanup, all N_{actual} verified non-degenerate)

M	Family	N_{actual}	a_{hole} (μm)	density (μm^{-2})	ghost margin
0.0	rectangular	11	2000	2.963e-07	inf
1.0	jittered lattice	20	1250	9.486e-08	inf

M	Family	N_{actual}	a_{hole} (μm)	density (μm^{-2})	ghost margin
2.0	jittered lattice	20	750	1.013e-07	inf
3.0	jittered lattice	20	750	1.013e-07	inf
4.0	archimedean spiral	50	600	2.308e-07	inf
5.0	archimedean spiral	50	600	2.308e-07	inf
6.0	jittered lattice	20	500	1.047e-07	inf
8.0	jittered lattice	20	400	1.061e-07	inf
global	square	9	2000	2.500e-07	inf

4.2 Metric 2

At the current tested spacing resolution, Metric 2's Pareto front collapses to a single point at every M (verified genuine via a synthetic-data control that confirmed the dominance logic itself produces multi-point fronts when the underlying data has a live density/margin trade-off). This reflects that safety in the present dataset is largely binary — trivially-safe (zero active pairs) versus clearly unsafe — rather than a graded trade-off region. Widening the tested spacing range toward the safety boundary is the natural next probe if a live Pareto trade-off is worth resolving explicitly.

5 Honest Scope Notes

- Hole diameter is fixed from the acuity-optimal chart, not searched — this study answers "which geometry, at the size already known to be acuity-optimal," not "which size and geometry jointly."
- No brightness/transmission constraint is applied in this pass, by explicit decision, so that reproducibility/manufacturability considerations for the two stochastic families could be observed without being pre-filtered by an unrelated gate.
- This study uses the validated *closed-form* ghost and vignetting formulas only — not the full multi-hole wave-optics (ASM) engine. A subsequent confirmation pass on the Metric 1/Metric 2 finalists using the full validated propagation engine is the recommended next step before any physical fabrication decision.
- `family_ranking.csv` (an earlier, independent output) was checked and confirmed to draw only on ghost/vignetting-margin columns, not density — it was not affected by the degeneracy defect described in Section 2.

6 Addendum: Phase 2 Full-Physics Confirmation

Following this report's own Section 6 recommendation, the Metric-1 finalists (Table 5) were propagated through `array_optimization`'s validated multi-hole ASM engine (imported directly, not copied) and their blur and ghost separation were measured directly by peak detection on the resulting retinal intensity field, rather than predicted from the closed-form formula. Full methodology, tolerances, and per-finalist detail: `array_geometry_lab_phase2_confirmation/CONFIRMATION_REPORT.md`.

Result: 9/9 finalists CONFIRMED – every finalist's structural ghost/vignetting prediction (zero or one active hole under the pupil, hence no resolvable ghost) was reproduced by the full propagation at every one of 11 tested gaze angles ($\phi = -25 \ldots 25°$, step 5°). The one finalist with a non-trivial case – `rectangular` at $M = 0$D, where two holes become simultaneously active at $\phi = \pm 5°$ – confirmed the corrected formula's own $M = 0$ boundary prediction ($\Delta r = s \cdot M/(P_0 - M) \equiv 0$ at $M = 0$, independent of pair distance s) exactly: the real propagation's demagnification factor was independently measured at exactly 0.0 at $M = 0$D, so the two illuminated holes' retinal copies coincide and resolve as one peak, not two.

One divergent finding, on the blur reference, not the finalists' own physics: for 3 of the 9 finalists (the ones using $a_{\text{hole}} \geq 1250\,\mu$m), `new_sweeps/blur_sweep.csv`'s stored single-hole $r_{EE,50}$ (computed at grid resolution $N = 2048$) differs from a fresh recomputation at $N = 4096$ (the corrected default already adopted elsewhere in this project) by 16–100% – confirmed by direct recomputation at both resolutions, not assumed. Diagnostically, three of these finalists with genuinely different apertures report the IDENTICAL $N = 2048$ value to six decimal places – real single-hole diffraction has no reason to give the same blur radius for different aperture sizes, so this is read as the signature of `array_psf_envelope.py`'s own documented Defect-1 mechanism (an aperture-independent grid artifact) dominating the $N = 2048$ measurement for apertures this large, not genuine physics. This does not change any affected finalist's safety classification (each is at $M = 0$D or $M = 1$D, where either the predicted separation is exactly zero or Phase 2 measured no resolvable second peak regardless), but `blur_sweep.csv` should be regenerated at $N = 4096$ before being trusted for other apertures $\geq 1250\,\mu$m.

finalist	a_{hole} **(μm)**	M **(D)**	**ghost/vign.**	**blur (arraying)**	**class.**
M0.0_rectangular	2000	0	agree	agree	CONFIRMED
M1.0_jittered_lattice	1250	1	agree	agree	CONFIRMED
M2.0_jittered_lattice	750	2	agree	agree	CONFIRMED
M3.0_jittered_lattice	750	3	agree	agree	CONFIRMED
M4.0_archimedean_spiral	600	4	agree	agree	CONFIRMED
M5.0_archimedean_spiral	600	5	agree	agree	CONFIRMED
M6.0_jittered_lattice	500	6	agree	agree	CONFIRMED
M8.0_jittered_lattice	400	8	agree	agree	CONFIRMED
Mglobal_square	2000	0	agree	agree	CONFIRMED

Table 3: Phase 2 full-physics confirmation summary, one row per Metric-1 finalist. Full numeric comparison: `phase1_vs_phase2_comparison.csv`.

7 Recommended Next Step

With all 9 Metric-1 finalists confirmed under full-physics propagation (Section 7), the remaining gap is between simulation and reality: no design in this study has yet been checked against a physically fabricated sample. Immediate next steps: (1) regenerate `new_sweeps/blur_sweep.csv` at N=4096 project-wide, not just for the 9 finalists spot-checked here; (2) select 2–3 finalists spanning the tested M range for laser-cut shim fabrication, prioritizing manufacturability margin over raw blur/density optimality; (3) measure blur, transmission, and gaze-angle ghosting on the fabricated prototypes using the project's established phone-camera afocal photography protocol, and compare directly against this report's and Section 7's predictions.